

Advanced Deep Learning Solutions for Automated Diagnosis of Solar Panel Issues

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Abstract – Solar energy stands out as a highly reliable renewable technology with global applicability. The growing integration of solar photovoltaic systems worldwide emphasizes the need for sustaining peak performance and maximizing energy output. Despite their widespread feasibility, solar panels face various vulnerabilities, including dust accumulation, snow, bird droppings, and other anomalies that can compromise module efficiency and overall energy production. Because of this cleaning solar panels has become a critical task. Leveraging the effectiveness of Convolutional Neural Networks (CNNs) in image classification, particularly for visual inspection tasks, this paper compares the results of CNN VGG16 and RESNET50 architectures in detecting faults in solar panels using thermal images. Kaggle's image dataset encompasses various fault types such as dust, snow, bird droppings, and physical and electrical damage on solar panel surfaces. The suggested VGG16 CNN model attains a noteworthy success rate of 95.4%, while RESNET50 achieves a comparatively lower accuracy of 83%. Notably, this paper distinguishes itself by expanding the scope of previous works, which typically focused on detecting three fault classes. In contrast, the proposed model successfully identifies six fault classes, using both architectures significantly advancing solar panel fault detection.

Keywords – Solar Photovoltaic systems, Fault detection, Convolutional Neural Networks (CNNs), VGG16 architecture, RESNET50, Fine-tuned.

I. INTRODUCTION

In light of the critical imperative to attain net-zero carbon emissions globally by 2050, the energy landscape is undergoing a transformative transition from conventional hydrocarbon power generation to renewable sources. Among the most promising renewable energies, wind, hydroelectric power, and solar stand out at the top. Photovoltaic energy, harnessed through solar panels, emerges as a clean and abundant resource with a low environmental impact. However, due to many reasons, solar panel deployment is susceptible to various operational issues, ranging from broken glass and dust accumulation, adversely affecting their efficiency.

Previous research works have delved into fault detection using intelligent algorithms, including convolutional neural networks. The authors applied the techniques of cnn to detect problems in photovoltaic cells. [1] The study provided a feature extraction and classification method on a deep two-dimensional (2-D) CNN. However, it's always been challenging to determine the most effective classifiers due to the dependence on various parameters such as input characteristics, network architecture, etc.

This experiment employs two distinct deep-learning architectures, namely VGG16 and RESNET50. The algorithm initially categorizes input thermal images into six different classes in both models. Subsequently, pre-trained VGG16 CNN and RESNET50 CNN models are imported. Following this, pooling, dropout, and softmax layers are incorporated into the models. Ultimately, fine-tuning is applied to enhance the performance of the models.

VGG16 and RESNET50 are convolutional neural networks widely employed for image classification. The key distinction between these two architectures lies in the number of layers; VGG16 comprises 16 layers, while RESNET50 features 50 layers. Both architectures have demonstrated state-of-the-art performance on various benchmark datasets, including ImageNet. They are favored choices due to their versatility, accuracy, and efficiency compared to other techniques. Moreover, they are suitable for transfer learning, a technique enabling the utilization of a pre-trained model to enhance the performance of a new model, especially when dealing with a small dataset for the new model. Considering these factors, our proposed work aims to advocate for an efficient and accurate fault detection system using the CNN VGG16 model designed to detect six fault classes.

II. LITERATURE SURVEY

Seung Heon Han et.al. [5] introduced a deep-learning approach for detecting faults in solar panels. The authors proposed a system that utilizes an unmanned aerial vehicle (UAV) equipped with a thermal camera and gps to capture thermal images and localize faults in solar panels. The deep learning model and improved version of You Only Look Once (YOLOv3-tiny), are employed for fault detection, and the information is transmitted to a remote server using long-term evolution for visualization.

Zhendong Huang et.al. [1] proposed an efficient method for fault diagnosis for solar panels using a convolutional neural network (CNN). This method involves a two-step solution: solar panel location and classification of defects. The edge detection component, named ESPED, is based on a lightweight CNN architecture and is designed for real-time applications.

Finally, the proposed approach is evaluated on infrared images collected from different photovoltaic plants, demonstrating its effectiveness in fault diagnosis. Mary Pa et.al. [8] designed a fault detection scheme for photovoltaic solar panels using a convolutional neural network (CNN). The proposed CNN model achieves a good accuracy of 91.2% for binary classification that is normal and faulty, and 88.6% for multi-classification of 4 classes i.e., normal, cracked, dusty, and shadowed. CNN surpasses a prior model's performance on the identical dataset, achieving a 16% improvement for binary classification and an 18.6% enhancement for multi-class classification. The research additionally investigates the consequences of decreasing the layer count and assesses the effectiveness of pre-trained models, revealing their constraints to this particular dataset. Himanshu Bendale et.al. [7] developed a deep learning-based methodology for solar panel fault detection using convolutional neural networks. The system processes thermal images captured by unmanned aerial vehicles equipped with infrared sensors. The CNN model achieves a high accuracy of 97.5% in detecting various types of faults, surpassing previous methods using principal component analysis and kullback-leibler divergence (KLD). The system considers voltage current values to predict the amount of energy loss due to defects. The proposed model demonstrates superior performance, handling multiple fault types and providing a robust solution for real-time solar panel maintenance.

Nikhil Prajapati et.al. [9] introduced a real-time method for detecting and identifying faults in photovoltaic modules using thermal images. The proposed approach employs a convolutional neural network-based algorithm, specifically You Only Look Once (YOLO), to detect and classify four types of faults: temporary hotspot, permanent hotspot, bypass diode, and crack/wear and tear. The study uses a dataset thermal image captured by a drone, and the results indicate reliable fault detection with a maximum mean average precision of 83.86%. However, the model's precision could be further improved with an increased dataset, especially for bypass diode faults. Monica Patil et.al. [6] proposed an automatic fault detection and localization method for solar photovoltaic (PV) systems. Utilizing MATLAB/Simulink and hardware experiments with a 3*3 solar panel configuration, the approach compares residual faults to a threshold value, effectively identifying and locating faults between lines within a solar rooftop string. The study focuses on cross-string and intra-string faults which are very challenging to detect with conventional protection devices in large-scale PV systems. Experimental results, including led-based visual indicators, demonstrate the method's accuracy and effectiveness in real-time fault monitoring PV systems.

II. PROPOSED METHODOLOGY

A. Dataset Acquisition

In this analysis of solar fault detection utilizing convolutional neural networks, the acquisition of a suitable dataset plays a pivotal role in the success of the entire research endeavor. The dataset serves as the foundation upon which the CNN model is trained and evaluated, enabling it to learn the intricate patterns and characteristics that distinguish faulty solar panels from good ones.

As illustrated in Fig. 1., the suggested methodology consists of 4 primary steps: vision-based data acquisition, feature extraction, and classification respectively.

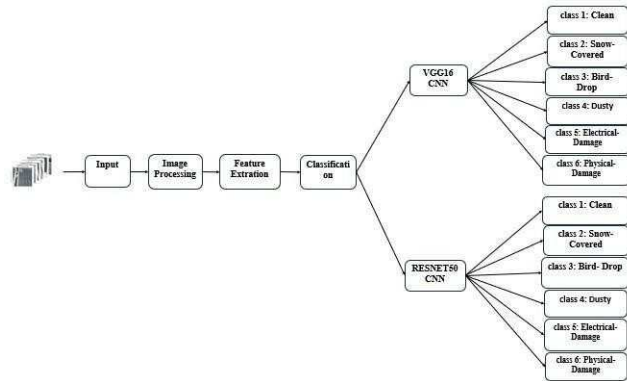


Fig. 1. Block diagram showing the steps involved in the proposed methodology.

The dataset used in this model is a Kaggle dataset which is publicly available under the name of "Faulty_solar_panel" which encompasses a comprehensive collection of thermal images of solar panels, capturing a wide range of faults. This diversity ensures that the trained cnn model is exposed to a variety of fault signatures, enhancing its ability to generalize to real-world scenarios.

The following six classes are considered for fault detection:

- Clean
- Bird-dropping
- Snow-covered
- Physical damage
- Electrical damage

B. Image Preprocessing

Resizing images is a fundamental preprocessing technique extensively utilized in computer vision applications, notably in tasks like solar fault detection employing Convolutional Neural Networks (CNNs). In such contexts, where consistency and standardization are critical, resizing images to uniform dimensions significantly enhances model effectiveness.

In the domain of thermal imaging of solar panels, inputs to fault detection models often exhibit size disparities due to various factors like camera specifications, distance from the target, or orientation.

These discrepancies can introduce undesirable noise and inconsistencies, potentially hindering the model's ability to learn relevant fault patterns. Standardizing all images to a specific dimension, such as 244x244 pixels, addresses these issues.

The resizing process entails scaling each image to fit within the specified dimensions, ensuring dataset uniformity. This uniformity mitigates distractions arising from variations in image size, enabling the model to concentrate solely on intrinsic fault patterns.

Moreover, standardizing image dimensions streamlines the training process by reducing computational complexity. Models trained on consistently sized images are more efficient and demand fewer computational resources during both the training and inference stages. Additionally, resizing images to a fixed dimension aids in generalization.

In summary, image resizing serves as a crucial preprocessing step, pivotal in improving the performance and robustness of machine learning models, particularly in computer vision tasks such as solar fault detection. By enforcing uniform image dimensions, this technique ensures the model's focus remains on relevant features necessary for accurate fault identification, resulting in more dependable and effective detection systems.

C. Feature Extraction

In the domain of machine learning, data splitting stands as a foundational technique utilized to ensure the efficacy of model training and evaluation. This method entails dividing the dataset into distinct subsets, each serving a specific role within the machine learning workflow. The primary objectives of data splitting encompass averting overfitting, enhancing model generalization, and accurately assessing model performance on unseen data.

Overfitting poses a prevalent challenge in machine learning, wherein a model excessively memorizes the training data instead of discerning and generalizing underlying patterns and relationships. When a model overfits, it demonstrates exceptional performance on the training data but struggles to generalize effectively to novel, unseen data instances. This phenomenon undermines the model's performance and reliability in real-world scenarios.

By employing data splitting, practitioners mitigate the risk of overfitting by restricting the model's access to the entire dataset during training. Typically, the dataset is partitioned into three primary subsets:

1. Training set: The training set comprises the largest segment of the dataset and is used to train the model parameters. Through respective learning from the training data, the model refines its parameters to minimize prediction errors.
2. Validation set: The training set serves as a bridge between the training and testing sets. The validation set is employed to monitor the model's performance during training and to fine-tune hyperparameters, preventing overfitting.

3. Testing set: The testing set remains untouched during training and is used exclusively to evaluate the model's final performance on unseen data. The testing set provides an unbiased assessment of the model's generalizability to real-world scenarios.

In the specific context of image training, data splitting assumes heightened significance. Without meticulous partitioning of the dataset, the model may inadvertently fixate on the specific image features or noise rather than extracting meaningful patterns. Through the utilization of training, validation, and test sets, practitioners can facilitate the training of models that adeptly discern the underlying features of images, consequently leading to enhanced performance and generalization on unseen data.

Using the below strategy, the dataset is divided as 80% to the training set and the rest 20% for validation. The split ensured that the model received sufficient training data while also having a dedicated set for monitoring overfitting and tuning hyperparameters.

The total number of images taken for this paper is as follows:

1. 194 images for class-1 i.e., Bird-drop
2. 194 images for class -2 i.e, Clean
3. 191 images for class-3 i.e., Dusty
4. 104 images for class-4 i.e, Electrical-damage
5. 70 images for class-5 i.e, Physical-Damage
6. 124 images for class-6 i.e., Snow-Covered

Here in the first model, pre-trained VGG16 is employed as the base model for fault detection, and in the second model pre-trained RESNET50 is employed. These architectures comprise a series of convolutional layers, pooling layers, and fully connected layers, each with its own set of parameters.

The convolutional layers in the models extract features from the input thermal images. These features represent the patterns and textures within the images that are relevant for fault detection. Each convolutional layer has a set of learnable parameters, including the weights and biases associated with its filters. The learnable parameters in this layer are weights and biases. The total number of convolutional layers present in this model is thirteen (13) with a stride size of 1.

The pooling layers reduce the spatial dimensions of the feature maps while preserving the extracted features by using biases as the learning parameters. This dimensionality reduction helps to control the computational complexity of the model and prevents overfitting. Each pooling layer has a set of parameters that determine the pooling size and stride. The number of pooling layers present here is five (5) with a stride size of 2. The fully connected layers in the model integrate the extracted features and learn a nonlinear mapping to the output classification labels. These layers also have a set of learnable parameters, including the weights and biases associated with the connections between neurons. and they are three (3) in number.

TABLE I. OVERVIEW OF THE ARCHITECTURE – VGG16 CNN

Layer Type	Number of Layers	Stride Size
Convolutional layers	13	(1,1)
Global Average Pooling layer	1	Not applicable(global pooling)
Dropout layer	1	Not applicable(global pooling)
Fully connected layer	1	Not applicable

The architectural design comprises 13 convolutional layers, each utilizing a stride size of (1,1). Subsequently, a global average pooling layer and a dropout layer are incorporated. Furthermore, a sole fully connected layer is integrated within the architecture.

TABLE II. OVERVIEW OF THE ARCHITECTURE – RESNET50 CNN

Layer Type	Number of Layers	Stride Size
Convolutional layers	49	Varies
Global Average Pooling layer	1	Not applicable (global pooling)
Dropout layer	1	Not applicable (global pooling)
Fully connected layer	1	Not applicable

The architecture comprises 49 convolutional layers, each configured with diverse stride sizes. After these convolutional layers, a singular global average pooling layer is introduced, alongside a dropout layer aimed at regularization. Moreover, the architecture encompasses a solitary fully connected layer tailored for the purpose of final classification tasks.

During training, the parameters of the models are adjusted using an optimization algorithm i.e., Adam or stochastic gradient descent (SGD). The goal of using this optimization algorithm is to minimize the loss function, which measures the discrepancy between the model's predictions and the true labels. By iteratively updating the patterns and characteristics associated with various fault types in solar panels.

D. Classification

In the classification process, convolutional layers play a crucial role in extracting essential features from input thermal images. These features capture intricate patterns and distinct characteristics associated with different types of faults present in the images. As data traverses through convolutional, pooling, and fully connected layers, these features undergo gradual transformation and integration.

The final fully connected layer in the architecture holds significant importance as it serves as the cornerstone of classification. Here, the extracted features are synthesized and fused to create a comprehensive representation of the input image. This layer performs a weighted summation of these features, generating a vector of values corresponding to the probabilities assigned to each fault category. The use of the softmax activation function further enhances this process. By normalizing output values into a probability distribution, the softmax function ensures that the model's predictions accurately represent the likelihood of each fault category for the given input image. Essentially, it yields a probability distribution over fault categories, offering insights into the relative probabilities of different fault types within the image.

Through an iterative process of feature extraction, transformation, and classification, the model proficiently discerns and distinguishes between various fault types depicted in thermal images. This sophisticated mechanism enables the model to be informed decisions based on observed patterns, thereby facilitating precise fault detection and classification.

IV . RESULTS AND DISCUSSION

In the ultimate phase of the process, solar panels are conclusively identified as one of the six predefined classes. The classification hinges on the calculated probabilities associated with each class. If the probability for the “clean” class attains the maximum value among the alternatives, the corresponding solar panel is identified as a “clean solar panel”. Analogously, for the remaining classes, the solar panels are assigned to the category corresponding to the highest probability, thereby achieving accurate and definitive classification outcomes.

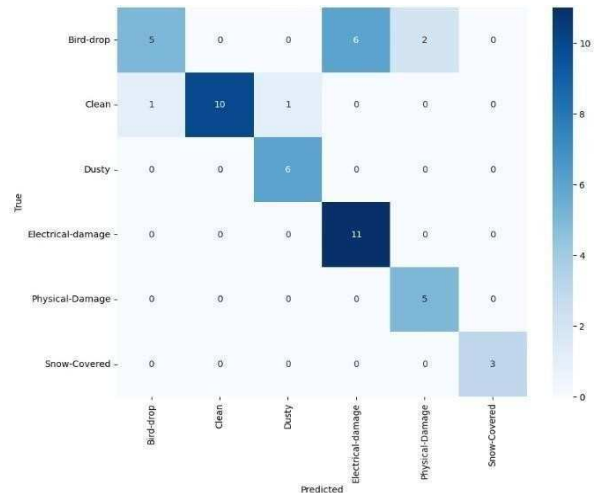


Fig.2. Confusion matrix

The experimental outcomes, as illustrated in Figure 3, offer insights into the efficacy of the developed model in identifying various faults in solar panels. The first image is appropriately identified as “clean”. The second image is characterized by a within the solar module. a layer of snow covering the panel. The third image exemplifies the model’s ability to detect bird droppings on a solar panel. In the fourth image, the model accurately identifies dust-related faults. The fifth image signifies the model’s proficiency in diagnosing electrical damage as the cause of a solar panel’s fault. Lastly, the sixth image depicts the model’s capability to recognize physical damage sustained by a specific solar panel.

In this way, the methodology for solar panel faulty detection is performed for six (6) classes.



Fig. 3. Display of solar panel detections of “Clean”, “Snow-Covered”, “Bird-drop”, “Dusty”, “Electrical-damage”, and “Physical-damage”.

V. CONCLUSION

In the realm of sustainable energy, where solar panels play a key role, ensuring their optimal functionality is of paramount importance. With more than 51.428 billion solar panels globally the need for an accurate and efficient fault detection system is evident. This paper, aimed at enhancing the efficacy of solar panel fault detection, addresses a critical aspect of renewable energy infrastructure. By leveraging advanced technologies, specifically convolutional neural networks VGG16, and RESNET50 architectures, this paper introduces a sophisticated model for identifying various faults in solar panels.

The methodology involves a systematic approach, beginning with preparing and organizing a comprehensive dataset consisting of diverse instances of faulty solar panels. Through the utilization of image processing techniques, the loaded images undergo preprocessing and feature extraction, laying the groundwork for the subsequent CNN-based analysis. On comparing the results, the VGG16 architecture could give a better accuracy of 95.4% than RESNET50 which could only give 83% with the same dataset. One of the distinguishing features of this model is its ability to categorize faults beyond a binary classification, extending its proficiency to recognize specific types of faults, including bird-dropping, dust accumulation, physical damage, electrical damage, snow coverage, and clean panels. This nuanced approach provides valuable insights for maintenance and intervention strategies in solar panel installation.

Looking forward, integrating intricate techniques like GoogLeNet, and Xception presents exciting opportunities for further refinement and increased accuracy. The implications of this paper extend beyond routine maintenance, offering a foundation for proactive fault mitigation in solar energy systems. As the world increasingly relies on renewable energy sources, this paper contributes significantly to the reliability and sustainability of solar power infrastructure studies.

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